

Kamrun Nahar Mily

PhD Candidate, Department of Mathematics, University of Houston

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Education

- **Ph.D. in Applied Mathematics** Spring 2022 – Expected Spring 2027
University of Houston, Texas, USA
 - Advisor: Dr. Ming Zhong
 - Research focus: Agent-Based Modeling for Crowd and Crime Dynamics
- **M.S. in Applied Mathematics** 2025
University of Houston, Texas, USA
 - GPA: 3.77/4.00
- **M.S. in Applied Mathematics** 2016 – 2017
University of Dhaka, Bangladesh
 - GPA: 3.97/4.00
 - A.F. Mujibur Rahman Gold Medal — First Class First
- **B.S. in Mathematics** 2010 – 2015
University of Dhaka, Bangladesh
 - CGPA: 3.49/4.00
 - Project: A Study of Differential Equations in Population Biology.

Teaching Experience

- **Instructor of Record** Spring 2026
University of Houston
 - Design and deliver a computation-driven linear algebra course integrating theoretical concepts with MATLAB-based applications.
 - Develop assignments, exams, and coding projects emphasizing conceptual understanding and problem solving.
 - Mentor students through office hours and support diverse learning needs.
 - Trained incoming Instructors of Record for Fall 2026, sharing course materials, assessment questions, and guidance on course delivery.

Spring 2026: MATH 2131 – Linear Algebra with MATLAB
- **Graduate Teaching Assistant** Spring 2022 – Present
University of Houston
 - Led weekly recitation and laboratory sessions for Calculus I–II, serving 80+ students per semester.
 - Conducted extended office hours and individualized tutoring for students with diverse mathematical backgrounds.
 - Guided students through proofs, problem-solving techniques, and exam preparation.
 - Collaborated with faculty on grading, exam design, and development of supplementary course resources.

Spring 2022: MATH 3331 – Intermediate Differential Equations (grader and tutor)
 Summer 2022: MATH 2412 – Calculus II (Teaching Lab and tutor)
 Fall 2022: MATH 1431 – Calculus I, MATH 3331 – Intermediate Differential Equations (grader and tutor)
 Spring 2023: MATH 2318 – Linear Algebra (grader and tutor)
 Summer 2023: MATH 2318 – Linear Algebra (grader and tutor)
 Fall 2023: MATH 2318 – Linear Algebra (grader and tutor)
 Spring 2024: MATH 2318 – Linear Algebra (grader and tutor)
 Summer 2024: MATH 2413 – Calculus I (Teaching online and tutor)
 Fall 2024: MATH 3331 – Intermediate Differential Equations (grader and tutor)
 Spring 2025: MATH 2415 – Calculus III (grader and tutor), Math 2318 – Linear Algebra (grader and tutor)
 Spring 2025: MATH 2318 – Linear Algebra (grader and tutor and proctoring exams)
 Fall 2025: MATH 2413 – Calculus I (Teaching in the lab and tutoring), MATH 3331 – Intermediate Differential Equations
 Spring 2026: MATH 2131 – Linear Algebra (IOR), MATH 2305 – Discrete Mathematics (grader and tutor)
 Summer 2026: Cataloging CCS quizzes
 Fall 2026: MATH 3363 – Introduction to Partial Differential equations (grader and tutor)

• Lecturer

Aug 2017 – Jul 2021

Primeasia University, Dhaka, Bangladesh

- Primary instructor for up to six undergraduate mathematics courses per semester, cohorts of up to 80 students.
- Designed comprehensive syllabi, delivered interactive lectures, and developed innovative assessment strategies.
- Coordinated tutorial sessions and mentored undergraduates in mathematical research projects.
- Served on the curriculum development committee for the mathematics program.

Courses taught: Differential Calculus; Integral Calculus; Business Mathematics; Ordinary Differential Equations; Complex Analysis; Numerical Methods; Linear Algebra

Teaching Development

- TYRIT (Transfer Your Research Into Teaching) Course Design Workshop, Graduate School, University of Houston.
- Workshop on Presentation, Goal Setting, and Professional Development, Life Carnival Foundation.
- Participants of three minute thesis competition organised by the graduate school of UH
- Volunteer Tutor, Houston-area mathematics outreach programs, 2022 to present.

Research Interests

- Agent-based modeling of crowd dynamics and crime pattern analysis
- Pattern formation and parameter learning in complex systems
- Data-driven and kinetic models of collective social behavior
- Numerical optimization, finite volume methods, and scientific computing

Publications

Published

- **Data-driven Learning to Enhance a Kinetic Model of Distressed Crowd Dynamics.**
 Daewa Kim, Demetrio Labate, **Kamrun N. Mily**, and Annalisa Quaini.
Mathematical Models and Methods in Applied Sciences, 35(7):1609–1636, 2025.
 DOI: 10.1142/S021820252540007X. Preprint: arXiv:2411.12974.
- **A Finite Element Framework for Simulating Residential Burglary in Realistic Urban Geometries.**
 Baoli Hao, **Kamrun N. Mily**, Annalisa Quaini, and Ming Zhong.
Mathematical Models and Methods in Applied Sciences, 36(5):1019–1050, 2026.
 Preprint: arXiv:2508.11055.

Under Review

- **Crime Hotspot Dynamics in Residential Burglary Models with Police Response.**

Baoli Hao, Kamrun N. Mily, Annalisa Quaini, and Ming Zhong.

Submitted, 2026. Preprint: arXiv:2605.17709.

Conference and Seminar Talks

- **Minisymposium Talk**, SIAM TX–LA Sectional Conference, University of Texas at Austin, Sep 2025.
- **Graduate Student Paper Presentation (GSPP)**, AMS and SIAM Student Chapters, University of Houston, 2026.

Poster Presentations

- **Data-driven Learning to Enhance a Kinetic Model of Distressed Crowd Dynamics**
CBMS Conference, Department of Mathematics, University of Houston, Dec 2025.

Research Training

- ChAMELEON Summer School: advanced training in computational mathematics, University of Houston, 2025.
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- CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning, University of Houston, 2025.

Leadership & Service

- **Treasurer** Spring 2024 – Spring 2025
Association for Women in Mathematics (AWM) Chapter, University of Houston
 - Managed financial records and transactions for chapter activities.
 - Coordinated fundraising initiatives supporting student professional development.
- **Communication Chair** Fall 2022 – Fall 2023
Association for Women in Mathematics (AWM) Chapter, University of Houston
 - Developed communication strategies and maintained the chapter website.
 - Organized networking events and professional development workshops.
- **Co-organizer, Minisymposium on Agent-Based Modeling** Sep 2025
SIAM TX–LA Sectional Conference, University of Texas at Austin
- **Peer Reviewer** 2024 – Present
Graduate Student Research Symposium

Technical Skills

- **Programming:** Python (advanced), MATLAB (advanced), R (intermediate)
- **Scientific Computing:** Agent-based and kinetic modeling, finite element methods, numerical optimization; NumPy, SciPy, Pandas, Matplotlib
- **Mathematical Software:** Mathematica, SageMath, COMSOL
- **Tools:** $\text{\LaTeX}$, Beamer, Jupyter Notebooks, Git/GitHub, Canvas LMS, Microsoft Office Suite

Honors & Fellowships

- **First Place, Three-Minute Thesis (3MT) Competition** 2025
AMS Student Chapter, University of Houston
- **Graduate Tuition Fellowship** 2021 – Present
University of Houston

- **Teaching Assistantship Award**
University of Houston

2021 – Present
- **Bangladesh Sweden Trust Fund Fellowship**
Ministry of Finance, Bangladesh

2021
- **A.F. Mujibur Rahman Gold Medal (First Class First, M.S.)**
University of Dhaka

2017